Jacopo Umberto Verga, PhD

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SUMMARY

Ph.D. Computational Scientist specializing in high-dimensional genomic data and reproducible data systems. Proven track record at the **Dana Farber Cancer Institute** in developing **workflows (Nextflow, Docker)** for drug discovery and biomarker identification. Experience **building and deploying** production-grade **AI/ML systems for Drug Discovery** and Oncology. Expert at translating complex biological questions into scalable computational solutions and collaborating with cross-functional clinical teams to drive data-driven insights.

CORE SKILLS

AI & Engineering: GenAI (LLMs, RAG, Agents), Deep Learning, XGBoost, GPU Optimization (CUDA), Python, R. **Deployment & Infrastructure:** Nextflow (Reproducible Workflows), Docker (Containerization), GCP/AWS, CI/CD, HPC, Linux/Unix. **Life Sciences Domain:** Target Identification, Patient Stratification, Single-cell and spatial Omics, Clinical Trial Data, Immunology. **Operational:** Stakeholder Management, Project Scoping, Regulatory Compliance (GCP/Human Research certified), Technical Mentorship.

WORK EXPERIENCE

Broad Institute of MIT and Harvard/Dana Farber Cancer Institute — Boston, MA
Visiting Researcher, Medical Oncology — 02, 2024 – 11-2025

- **Directed the computational strategy** for multi-modal oncology research, bridging the gap between raw spatial data and actionable clinical biomarker insights.
- **Collaborated with clinicians and experimentalists** to design high-quality datasets and implement rigorous QC and statistical frameworks to ensure data correctness for translational research.
- **Engineered statistical frameworks for patient stratification** in CAR-T therapy, contributing to drug discovery efforts through data-driven biomarker evaluation.
- **Managed large-scale genomic data infrastructure on GCP**, leveraging Python and Docker to build reproducible, production-ready analysis environments for 400+ samples.
- **Drove significant workflow efficiencies**, achieving a 90% reduction in processing time through GPU optimization and ML-based automated image denoising.
- **Collaborated with StandardBiotools** to integrate 3D and GPU acceleration into imaging workflows, aligning technical milestones with strategic industry objectives.

University of Galway — Galway, Ireland
Researcher — 09, 2021 – 11, 2025

- **Engineered and deployed production-grade** bioinformatics pipelines using **Nextflow** and **Docker**, automating multi-omics data integration and ensuring full reproducibility for hematological research.
- Identified novel therapeutic targets and biomarkers through high-dimensional analysis of multi-omics data, with top candidates currently undergoing ***in vitro* experimentation** for functional validation.
- **Co-developed miRKatAI**, an Agentic AI system utilizing LLMs, **SQL querying**, and **ground search** to automate scientific literature mining and structured data extraction, significantly reducing manual research time.
- Applied **Deep Learning** for complex batch correction and statistical modeling to harmonize large-scale datasets, establishing rigorous evaluation benchmarks to ensure data quality and integration integrity.
- **Managed AWS cloud infrastructure** and mentored junior scientists in statistical programming and clean code practices, fostering a culture of maintainable and audit-ready software development.

University of Bern - Universität Bern — Bern, Switzerland
Visiting Researcher — 01, 2025 – 02, 2025

- Performed **Quality Control** and analysis of long-read **scRNA-seq** data, applying AI methods to unravel complex disease biology mechanisms.

CloudCix — Cork, Ireland
Tester — 01, 2023 – 06, 2023

- Benchmarked **Deep Learning** model training performance on novel GPU cloud infrastructure, utilizing NVIDIA CUDA Toolkit to optimize computational efficiency.

Ancona University Hospital — Ancona, Italy
Research Assistant, Immunology — 07, 2019 – 09, 2021

- **Analyzed clinical patient records and genetic data** for Immunology cohorts, contributing to personalized medicine and **clinical risk evaluation** reports.
- **Conducted comprehensive safety and efficacy analyses for CLS Behring** IV products.
- Collaborated with physicians to translate complex biological data into **clinical insights** for patient-centric research.

Queen's University Belfast — Belfast, UK
Intern — 02, 2020 – 08, 2020

- Executed **virtual screenings** of 500+ **small molecules**, utilizing molecular dynamics and protein-protein interaction networks.

EDUCATION

University of Galway Galway, Ireland
PhD, Genomics Data Science 09, 2021 – 11, 2025
Thesis: Computational methods for therapeutic target identification in hematological malignancies.
Relevance: Developed novel algorithms for drug discovery and multi-omics integration, directly applicable to identifying therapeutic targets in complex diseases.

Marche Polytechnic University Ancona, Italy
Master's Degree, Molecular and Applied Biology 2018 – 2021
Grade: 110L/100

CERTIFICATIONS

Google: GenAI Intensive Course (**04/2025**)
NVIDIA: Building Transformer-based NLP Applications (**11/2024**)
NVIDIA: Fundamentals of Deep Learning (**10/2024**)
CITI Program: Good Clinical Practice (GCP) for Clinical Research (**02/2024**)
CITI Program: Human Research (**02/2024**)
University of Galway: Project Management in the Real World (**11/2024**)
University of Pavia: Intensive School for Advanced Graduate Studies in Machine Learning (**09/2020**)

AWARDS

Thomas Crawford Hayes Travel Fund winner, **€2300**.
Longhack & VitaDAO 2023 hackathon Winning team, Team Leader, **€1000 + 1000 VITA-Coin**
1st prize for the Decisive Data Podium Session at the CSE Research Day, **€200**

PROJECTS

miRKatAI: Agentic AI for miRNA research, Google Gen AI, SQL, Prompt engineering, GroundSearch, Streamlit
Breast Cancer Chatbot: Assistant to analyze clinical/molecular data, Bioinformatics Workflows, AI Agents, LLM, Autoencoders
AI Maria: Fine-tuned LLM for Myeloma Research Assistant, Transformers, Fine tuning, RoBERTa

LANGUAGES

Language: English (C1 - Full Professional Proficiency), Italian (Native), Spanish (B2)

PUBLICATIONS

[1] Karen Guerrero-Vazquez et al. "miRKatAI: An Integrated Database and Multi-agent AI system for microRNA Research". In: (2025). DOI: 10.48550/ARXIV.2508.08331. URL: https://arxiv.org/abs/2508.08331.

[2] **Verga, Jacopo U**. and Konishi, Yoshinobu and Cordas Dos Santos, David M. et al. "Bone Marrow Spatial Signatures Predict Survival Outcomes and Toxicities after CAR-T Therapy in Patients with Relapsed/Refractory Multiple Myeloma". In: *Cancer Cell - Under review* (2025).

[3] Yoshinobu Konishi et al. "Bone Marrow Spatial Signatures Predict Survival Outcomes and Toxicities after CAR-T Therapy in Patients with Relapsed/Refractory Multiple Myeloma". In: *Blood* 144.Supplement 1 (Nov. 2024), pp. 590–590. ISSN: 1528-0020. DOI: 10.1182/blood-2024-204144. URL: http://dx.doi.org/10.1182/blood-2024-204144.

[4] **Verga, Jacopo U** et al. "NK Cell States in Multiple Myeloma: Pathways to Novel Immunotherapies". In: *Blood* 144.Supplement 1 (Nov. 2024), pp. 6857–6857. ISSN: 1528-0020. DOI: 10.1182/blood-2024-210297. URL: http://dx.doi.org/10.1182/blood-2024-210297.

[5] David Cordas dos Santos et al. "P-203 Ability To Perform Spatial Transcriptomics in FFPE Decalcified Bone Marrow Samples of Patients With Precursor Myeloma". In: *Clinical Lymphoma Myeloma and Leukemia* 24 (Sept. 2024), S154–S155. ISSN: 2152-2650. DOI: 10.1016/s2152-2650(24)02106-2. URL: http://dx.doi.org/10.1016/S2152-2650(24)02106-2.

[6] **Verga, Jacopo U** et al. "Genomic technology advances and the promise for precision medicine". In: *Therapeutic Drug Monitoring*. Elsevier, 2024, pp. 355–371. ISBN: 9780443186493. DOI: 10.1016/b978-0-443-18649-3.00007-0. URL: http://dx.doi.org/10.1016/B978-0-443-18649-3.00007-0.

[7] **Verga, Jacopo U** et al. "A Systems Biology Approach Reveals the Endocrine Disrupting Potential of Aflatoxin B1". In: *Exposure and Health* 16.2 (May 2023), pp. 321–340. ISSN: 2451-9685. DOI: 10.1007/s12403-023-00557-w. URL: http://dx.doi.org/10.1007/s12403-023-00557-w.

[8] Zachary Boswell et al. "In-Silico Approaches for the Screening and Discovery of Broad-Spectrum Marine Natural Product Antiviral Agents Against Coronaviruses". In: *Infection and Drug Resistance* Volume 16 (Apr. 2023), pp. 2321–2338. ISSN: 1178-6973. DOI: 10.2147/idr.s395203. URL: http://dx.doi.org/10.2147/IDR.S395203.

[9] Maria Giovanna Danieli et al. "Common Variable Immunodeficiency in Elderly Patients: A Long-Term Clinical Experience". In: *Biomedicines* 10.3 (Mar. 2022), p. 635. ISSN: 2227-9059. DOI: 10.3390/biomedicines10030635. URL: http://dx.doi.org/10.3390/biomedicines10030635.

[10] **Verga, Jacopo U** et al. "Integrated Genomic and Bioinformatics Approaches to Identify Molecular Links between Endocrine Disruptors and Adverse Outcomes". In: *International Journal of Environmental Research and Public Health* 19.1 (Jan. 2022), p. 574. ISSN: 1660-4601. DOI: 10.3390/ijerph19010574. URL: http://dx.doi.org/10.3390/ijerph19010574.

[11] George S. Hanna et al. "Contemporary Approaches to the Discovery and Development of Broad-Spectrum Natural Product Prototypes for the Control of Coronaviruses". In: *Journal of Natural Products* 84.11 (Oct. 2021), pp. 3001–3007. ISSN: 1520-6025. DOI: 10.1021/acs.jnatprod.1c00625. URL: http://dx.doi.org/10.1021/acs.jnatprod.1c00625.

[12] Veronica Pedini et al. "Incidence of malignancy in patients with common variable immunodeficiency according to therapeutic delay: an Italian retrospective, monocentric cohort study". In: *Allergy, Asthma & Clinical Immunology* 16.1 (June 2020). ISSN: 1710-1492. DOI: 10.1186/s13223-020-00451-z. URL: http://dx.doi.org/10.1186/s13223-020-00451-z.

[13] Maria Giovanna Danieli et al. "A Case of CVID-Associated Inflammatory Bowel Disease with CTLA-4 Mutation Treated with Abatacept". In: *Archives of Clinical and Medical Case Reports* 03.06 (2019). ISSN: 2575-9655. DOI: 10.26502/acmcr.96550124. URL: http://dx.doi.org/10.26502/acmcr.96550124.